

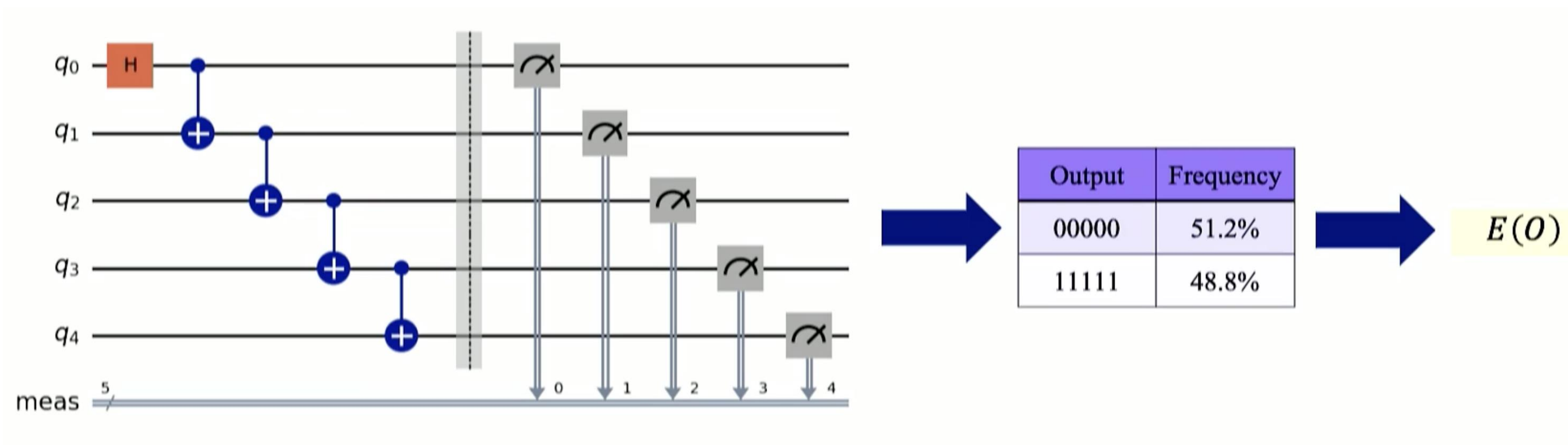
R2P Program Unit 3 - Lesson 2

Error Mitigation and Suppression

Ibrahim Shehzad
Quantum Algorithm Engineer
IBM Quantum

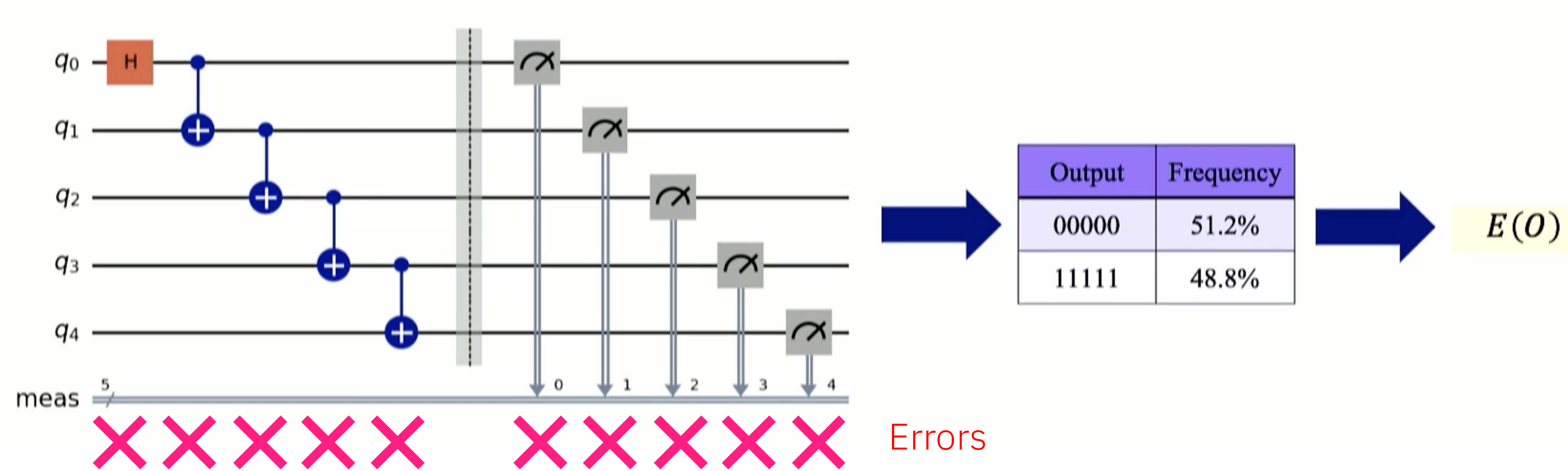
Motivation Behind Addressing Errors

Typically want to find expectation values in quantum computing.
Errors can be introduced every step of the way.



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Quick Glance: Error
Mitigation, Suppression,
Correction & Detection

Error Suppression: Working closer to the hardware and with physics to prevent errors before they happen (e.g. working on pulse sequences, etc.)

Error Mitigation: Using strategic classical techniques (software/algorithmic approaches) to work around errors without targeting the underlying hardware, often in post-processing

Error Correction: Using quantum codes with extra qubits to actively detect and correct errors

Error Detection: Using extra qubits to detect whether an error has occurred but not correcting it

Outline

1 Types of noise

Decoherence

Gate Errors

Cross-talk

Readout Error

Sampling Errors

2 Error Suppression and Error Mitigation

Dynamic Decoupling
(DD)

Twirling

Readout Error Mitigation

Zero Noise
Extrapolation (ZNE)

Probabilistic Error
Amplification
(PEA)

Probabilistic Error
Cancellation
(PEC)

Matrix-free
Measurement Mitigation
(M3)

Decoherence

- Information loss over time
 - High circuit depths increase noise
 - Decoherence can occur during idle cycles
- Transpilation methods can decrease noise
- [Dynamic Decoupling](#) specifically helps with idle cycles

ibm_torino OpenQASM 3

Details

133

Qubits

0.8%

EPLG

3.8K

CLOPS

Status:	● Online	Median CZ error:	4.384e-3
Total pending jobs:	43 jobs	Median SX error:	3.161e-4
Processor type ⓘ:	Heron r1	Median readout error:	1.770e-2
Version:	1.0.16	Median T1:	175.89 us
Basis gates:	CZ, ID, RZ, SX, X	Median T2:	134.83 us
Your instance usage:	0 jobs		

Gate Errors

- Caused by applying gates
- Two qubit gates are more noisy than single qubit gates
- Main source of error in high depth circuits
- Use good transpilation to decrease depth!
- Can be coherent or incoherent
- [Zero Noise Extrapolation \(ZNE\)](#) can be used to reduce noise

ibm_torino OpenQASM 3

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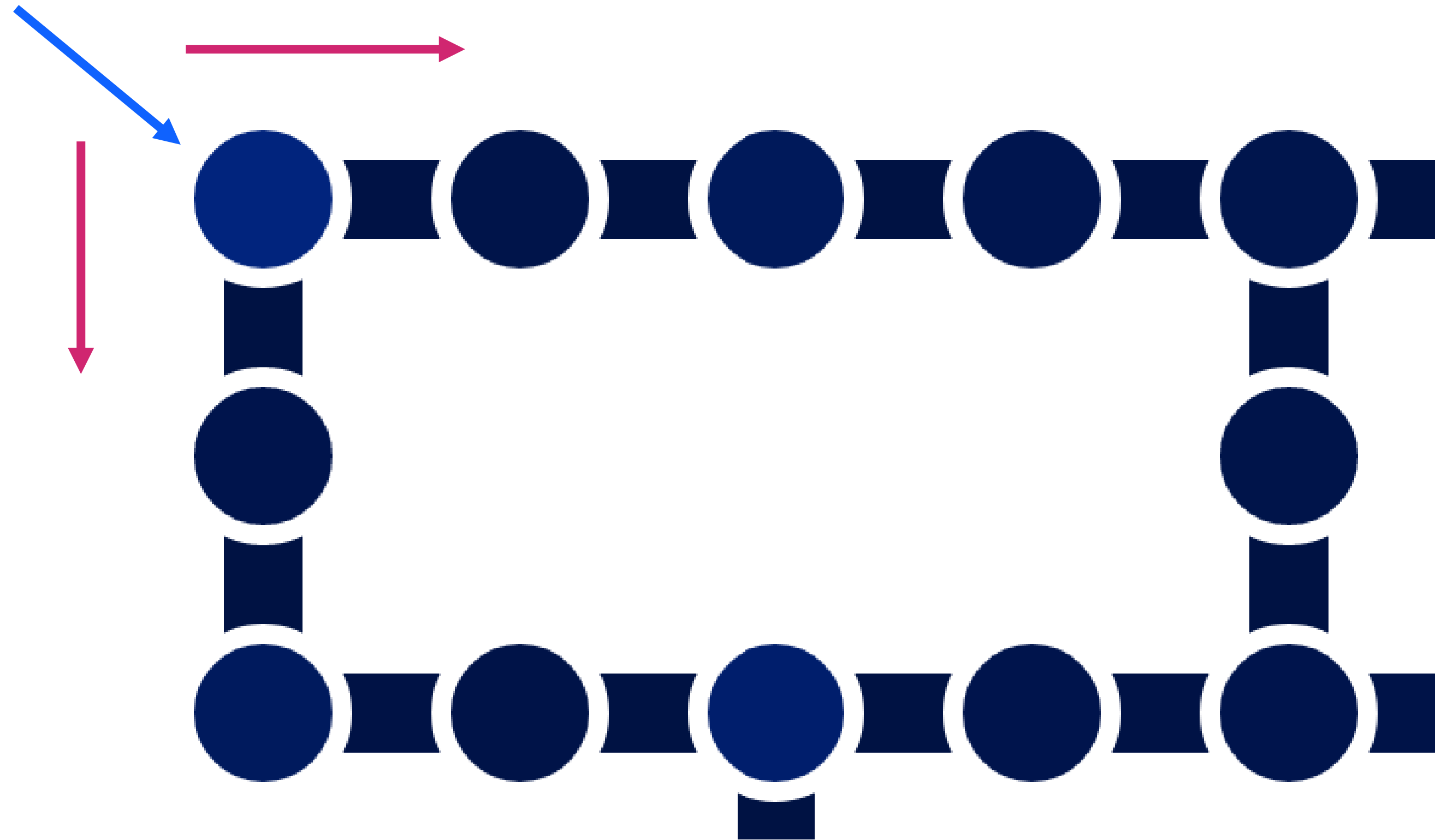
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Median T2: 134.83 us

Cross Talk

- Applying gates influences neighbouring qubits
- Hard to predict
- Mostly a matter of hardware
- Heron virtually eliminates cross-talk!



Readout Error

- Similar to classical noise
- Can be seen as a “noisy channel”
- Highest source of noise in low depth circuits
- [Twirled Readout Error eXtinction \(TREX\)](#) can be used to reduce noise

ibm_torino OpenQASM 3

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Shot noise

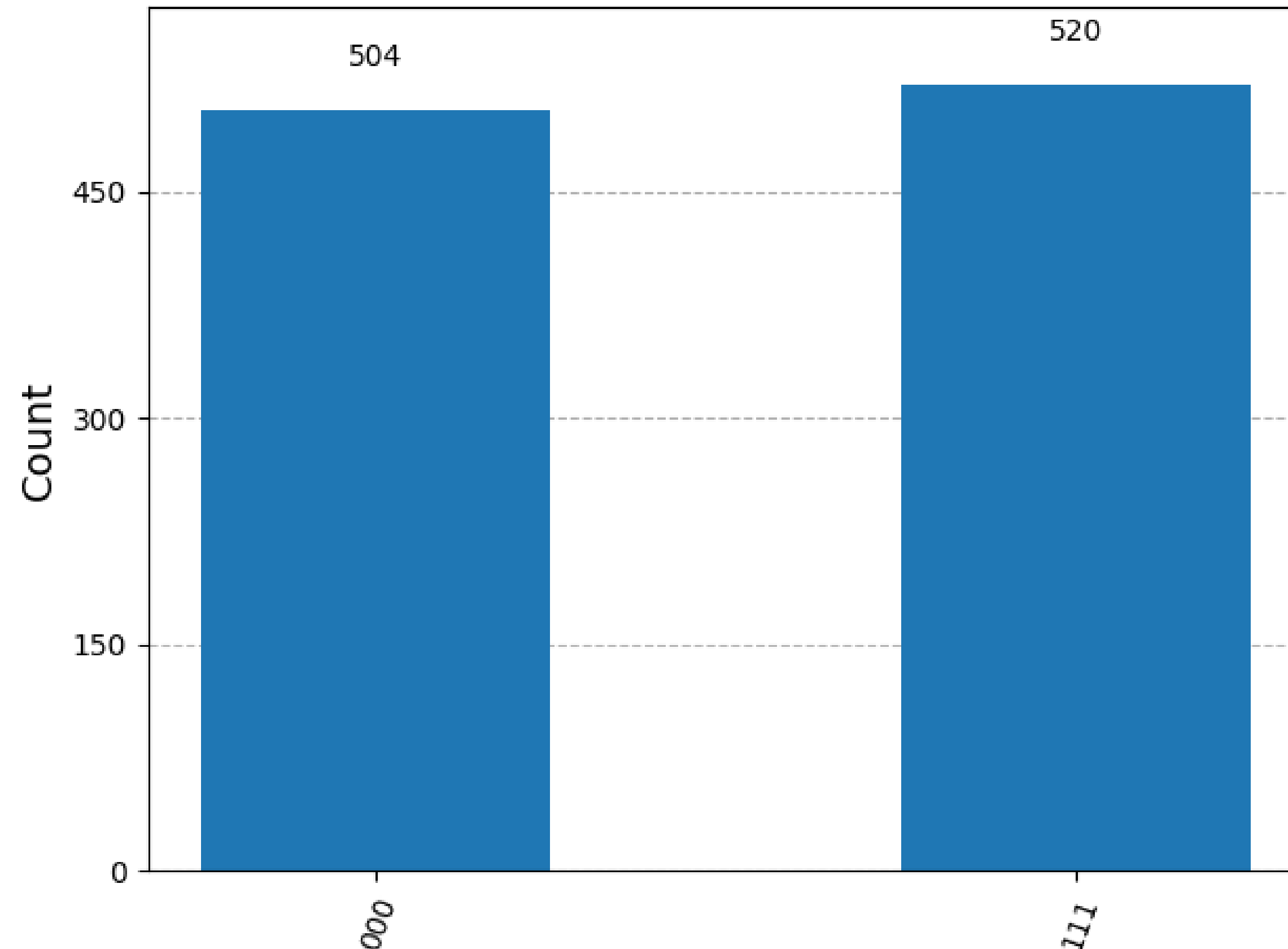
- Since a quantum experiment involves taking “shots” or sampling from an exponentially large distribution, you generally incur “sampling errors”.
- The sampling error decreases with more shots and vanishes as $n_{shots} \rightarrow \infty$

Consider GHZ state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

$P(0) = 1/2$
 $P(1) = 1/2$

In practice: $\rightarrow P(0) \approx P(1) \approx 1/2$



Key Relationships

- **Gate time $\ll T_1, T_2$:** Operations must be completed before decoherence (potentially thousands of operations!)
- **Error rates correlate:** Devices with better T_1/T_2 usually have lower gate errors
- **Temperature dependence:** All parameters improve at lower temperatures
- **Two-qubit gates** can be **ten times more error prone** than single qubit gates
- **$T_2 \leq T_1$:** Dephasing typically occurs faster than energy decay

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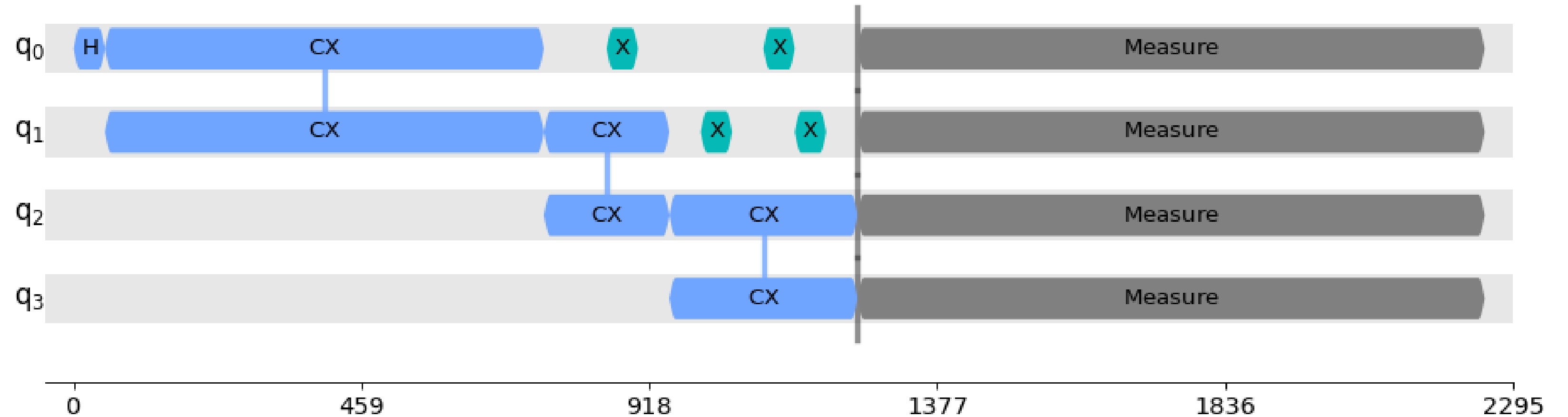
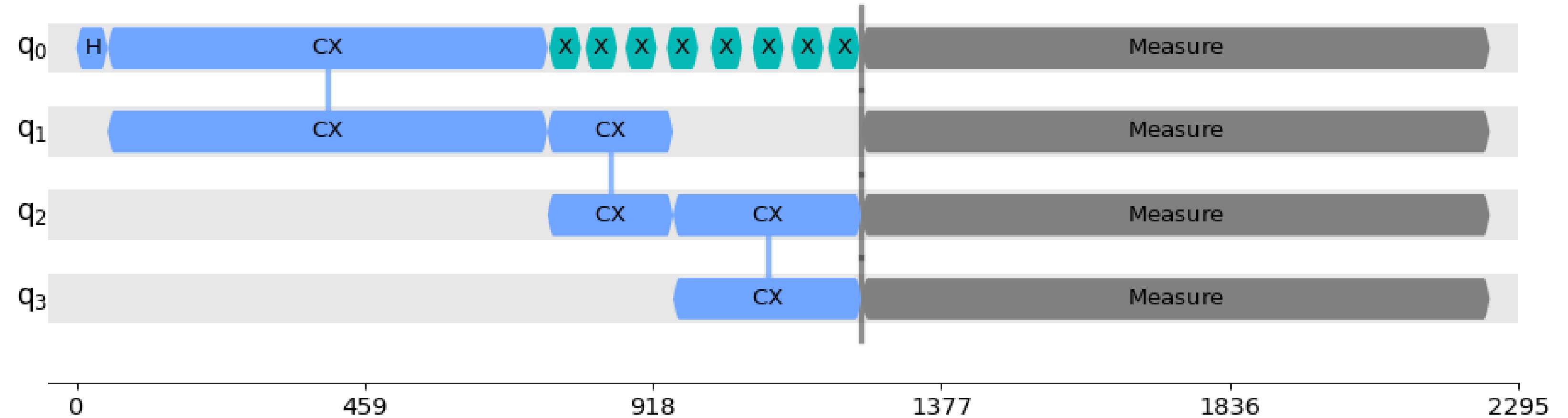
Probabilistic Error
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Measurement Mitigation
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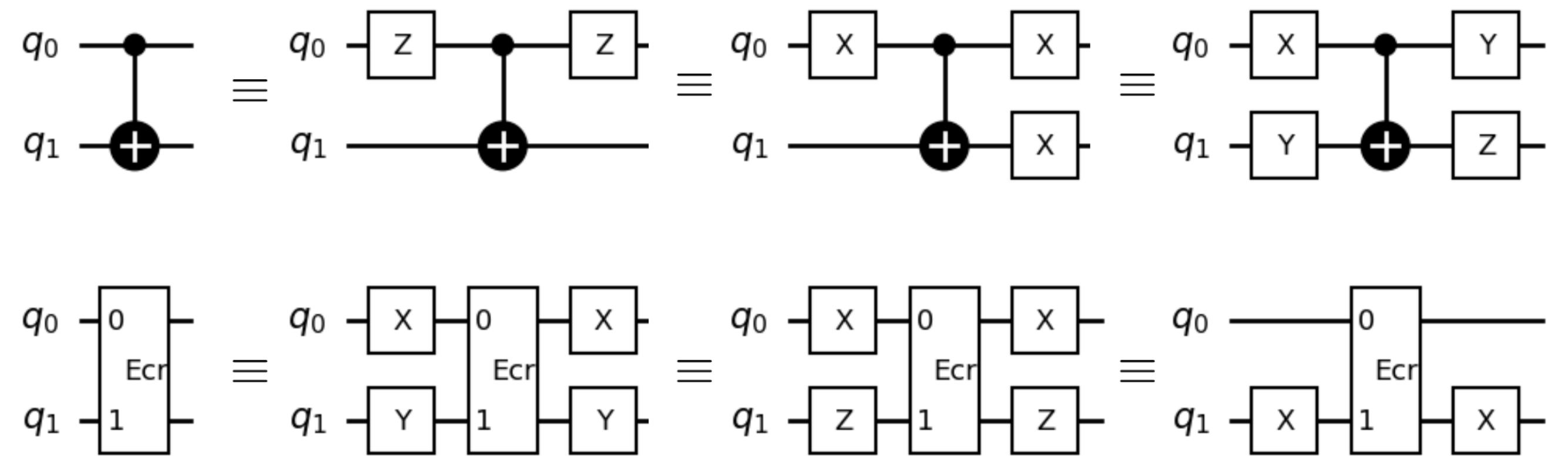
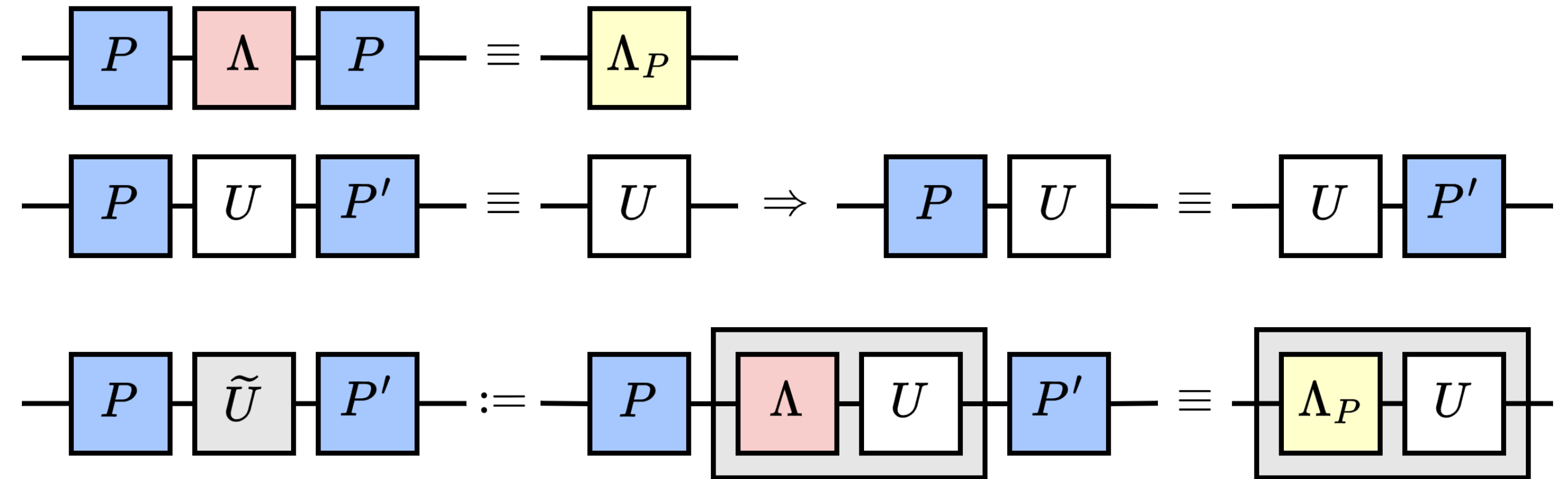
Dynamic Decoupling (DD)

- Idle cycles can cause decoherence.
- Having gates applied to qubits can help to mitigate this effect.
- The introduced gates add up to the identity.
- This is done in the transpiler.
- This is an option in Qiskit Runtime primitives.
- Works with both the Estimator and Sampler primitives



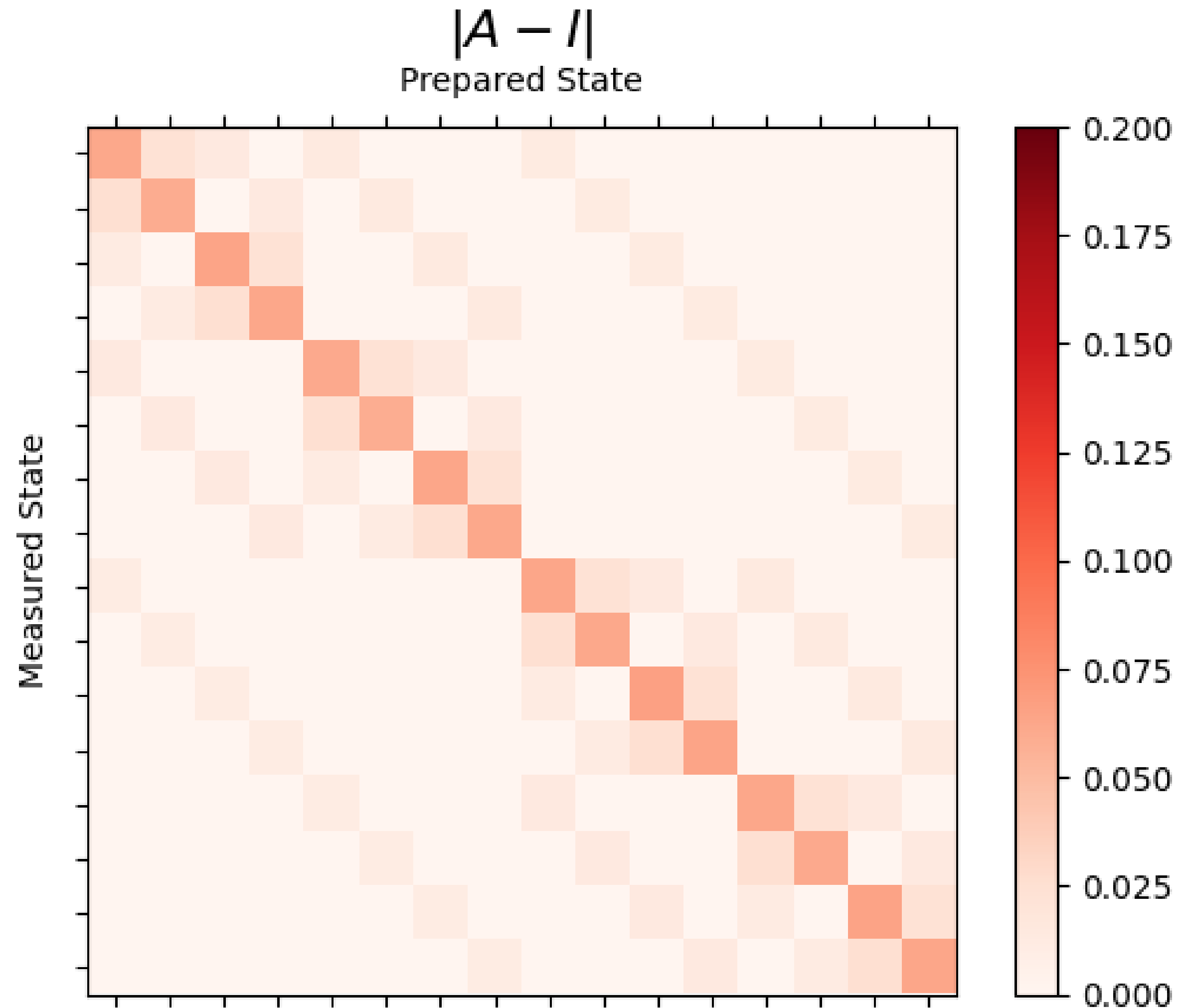
Twirling

- Also called randomized compiling.
- Used to convert arbitrary noise channels into other forms of noise.
- Helps when dealing with coherent noise.
- Helps in the extrapolation stage of ZNE by making noise increase more or less monotonically.
- Often exclusively used on two qubit gates.
- Qiskit runtime allows gate twirling and measurement twirling.
- Works with both the Estimator and Sampler primitives



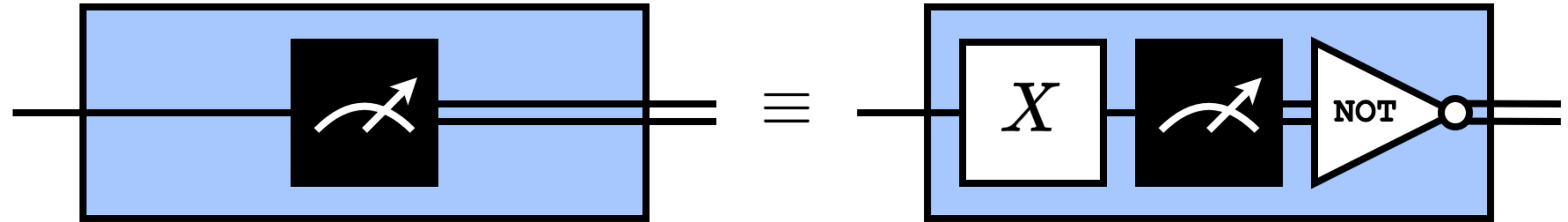
Readout Error Mitigation

- Readout errors cause the wrong states to be measured.
- This is the same as a classical “noisy channel”
- Readout error can be measured per qubit and then an error matrix can be constructed.
- The inverse matrix can be used for error mitigation.
- Can be found in [Qiskit Experiments](#).
- Not scalable to large circuits so runtime features [Twirled Readout Error eXtinction \(TREX\)](#) instead.



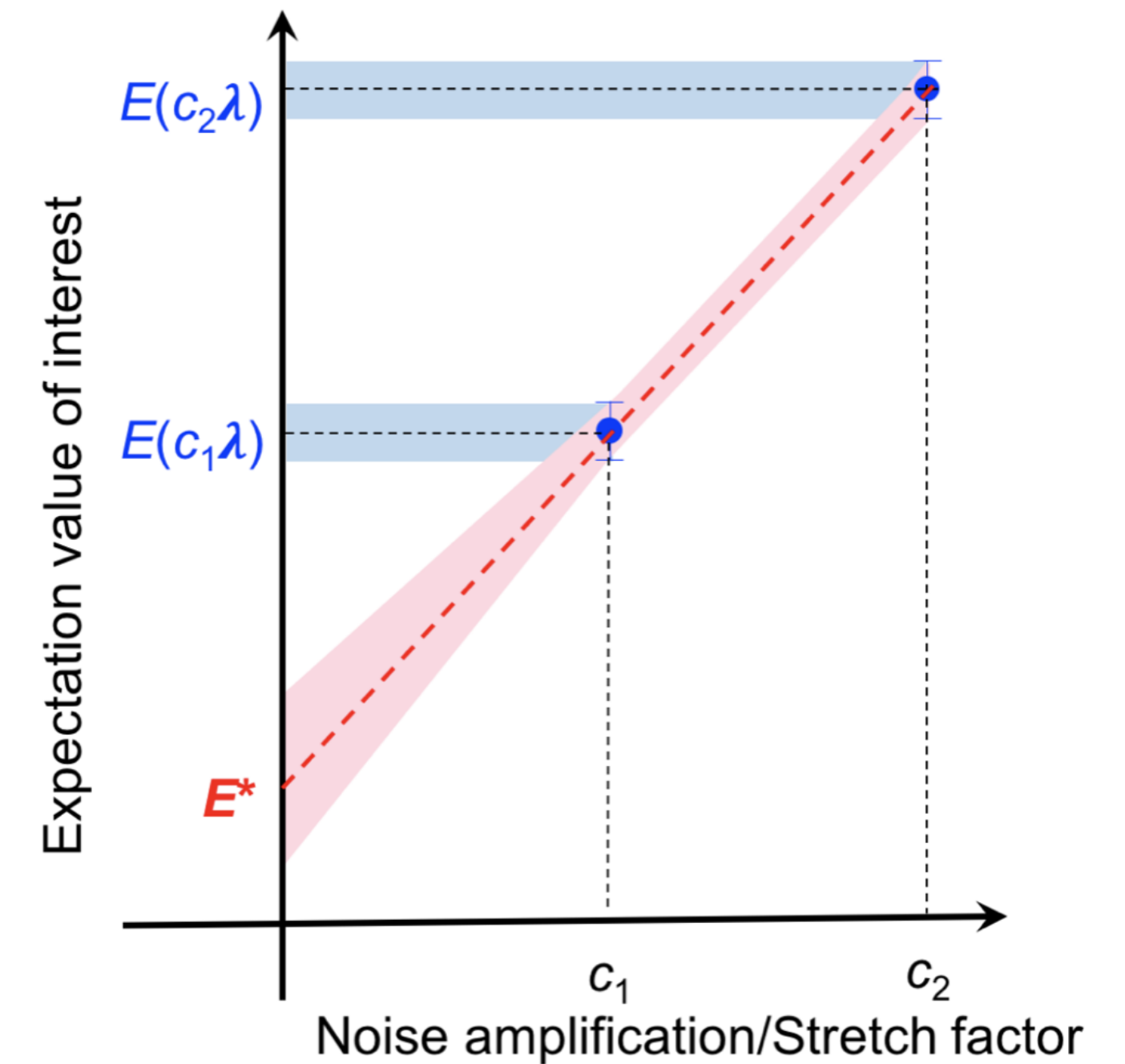
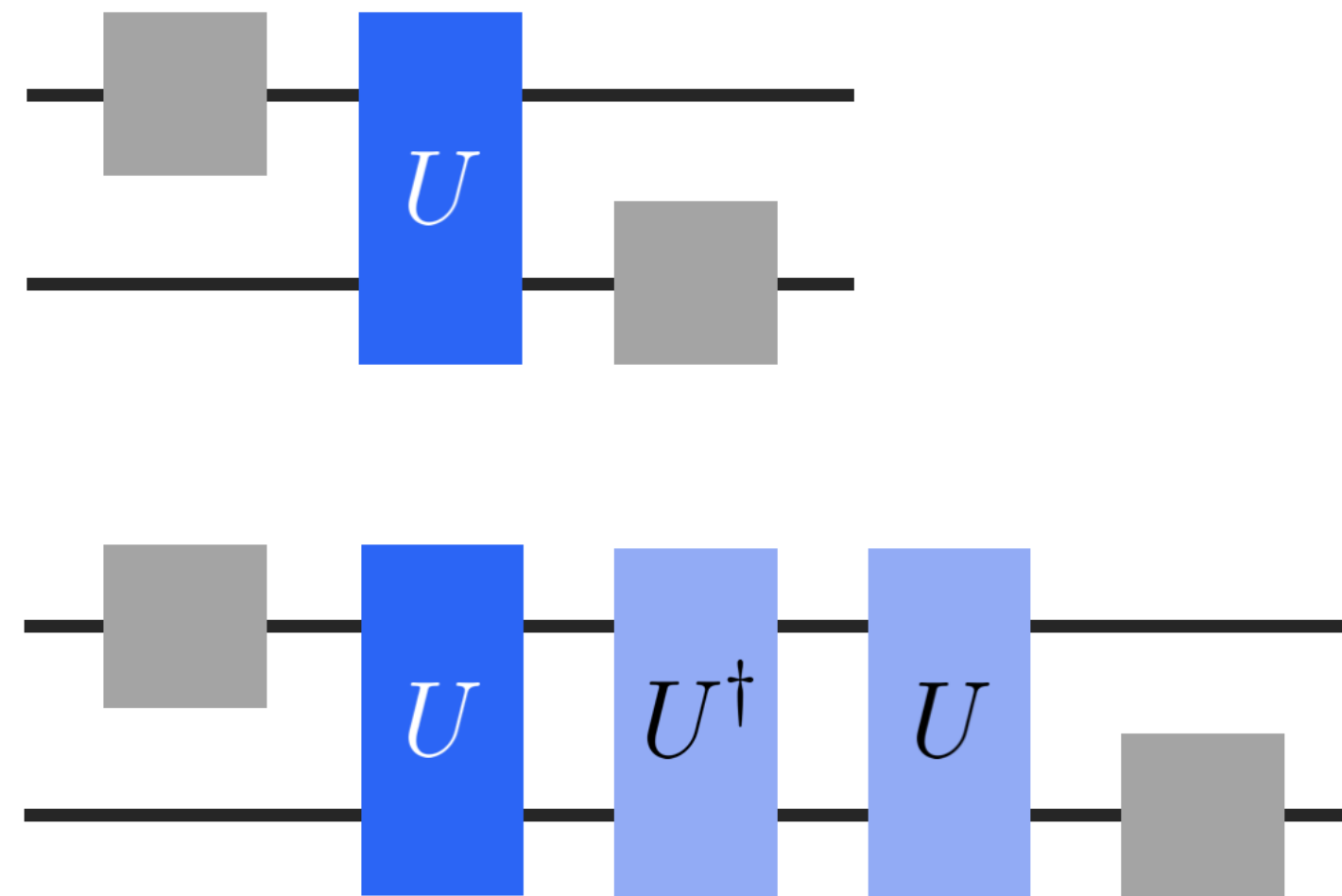
Twirled Readout Error eXtinction (TREX)

- Twirled Readout Error Extinction applies random bitflips before measurements.
- They are then undone in postprocessing.
- Efficient method thanks to diagonalizing readout-error transfer matrix.
- The calibration data is learned by benchmarking random circuits initialized in the zero state.
- Automatically applied in runtime with resilience level 1.
- Works with the Estimator primitive



Zero Noise Extrapolation (ZNE)

- Has two phases: Noise amplification and extrapolation.
- Twirling is used for preparation prior to noise amplification
- During noise amplification the original circuit is executed at different noise levels.
- During extrapolation the zero noise limit is inferred from the noisy expectation values.
- Automatically applied in runtime with resilience level 2.
- Works with the Estimator primitive

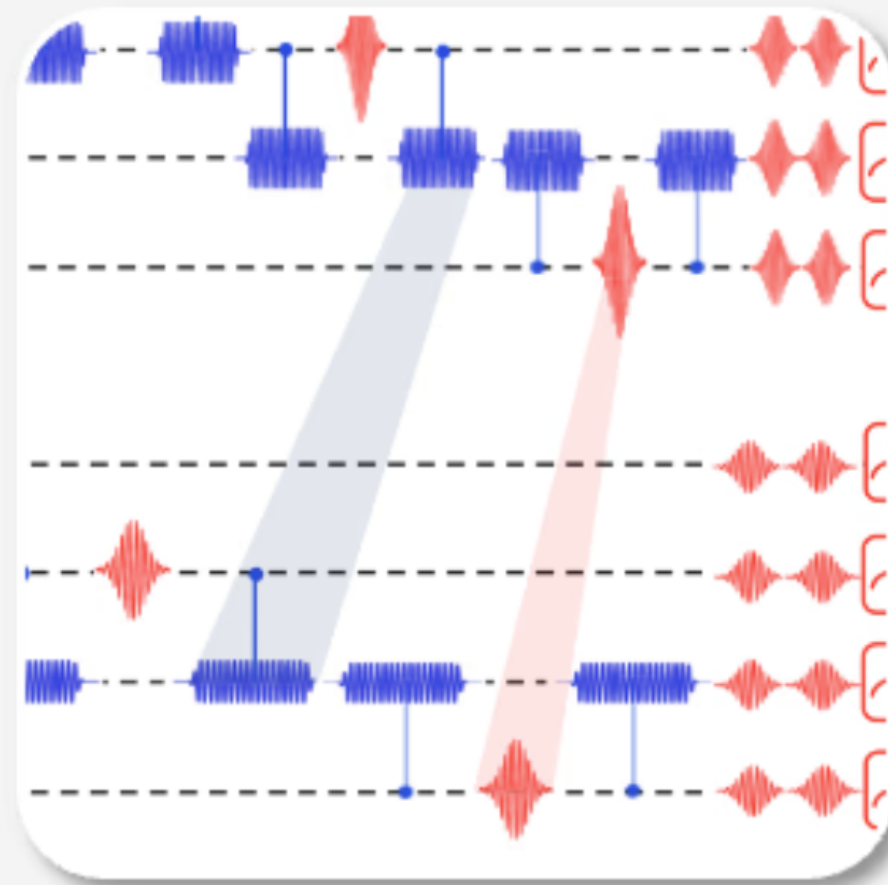


Noise Amplification

- Pulse stretching assumes gate noise is proportional to duration, which is typically not true. Calibration is also costly.
- Gate folding requires large stretch factors that greatly limit the depth of circuits that can be run.
- Probabilistic Error Amplification (PEA) can be applied to any circuit that can be run with native noise factor ($\lambda=1$) but requires learning the noise model.

Pulse Stretching

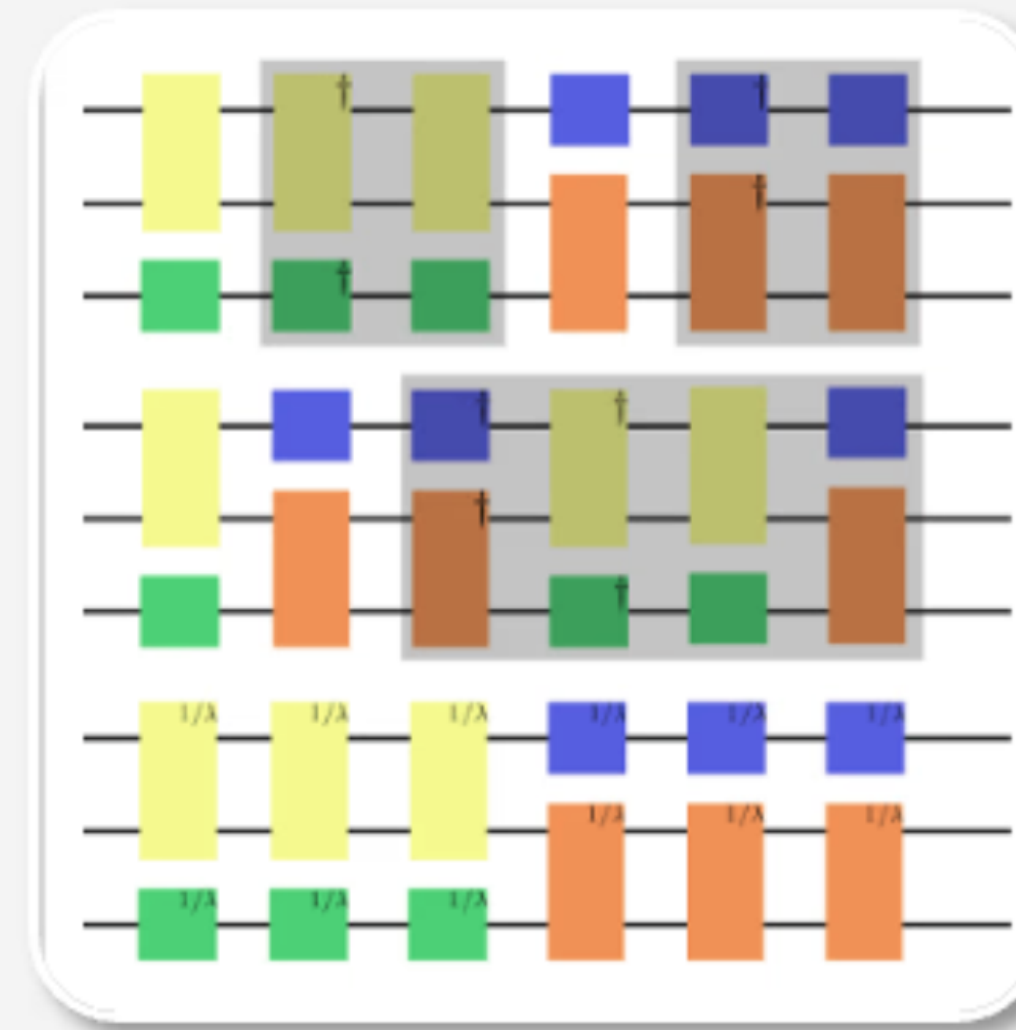
Scale pulse duration via calibration



Kandala et al. Nature (2019)

Gate Folding

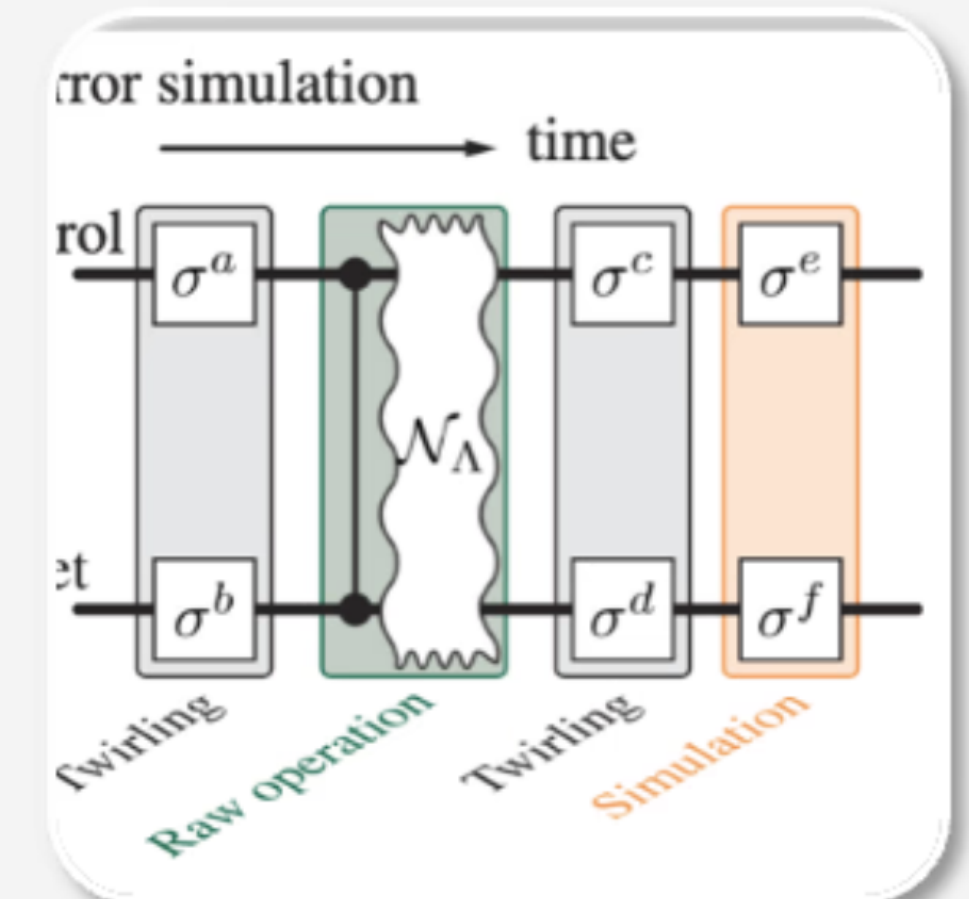
Repeat gates in identity cycles $U \mapsto U(U^{-1}U)^{\lambda-1}/2$



Shultz et al. PRA (2022)

Probabilistic Error Amplification

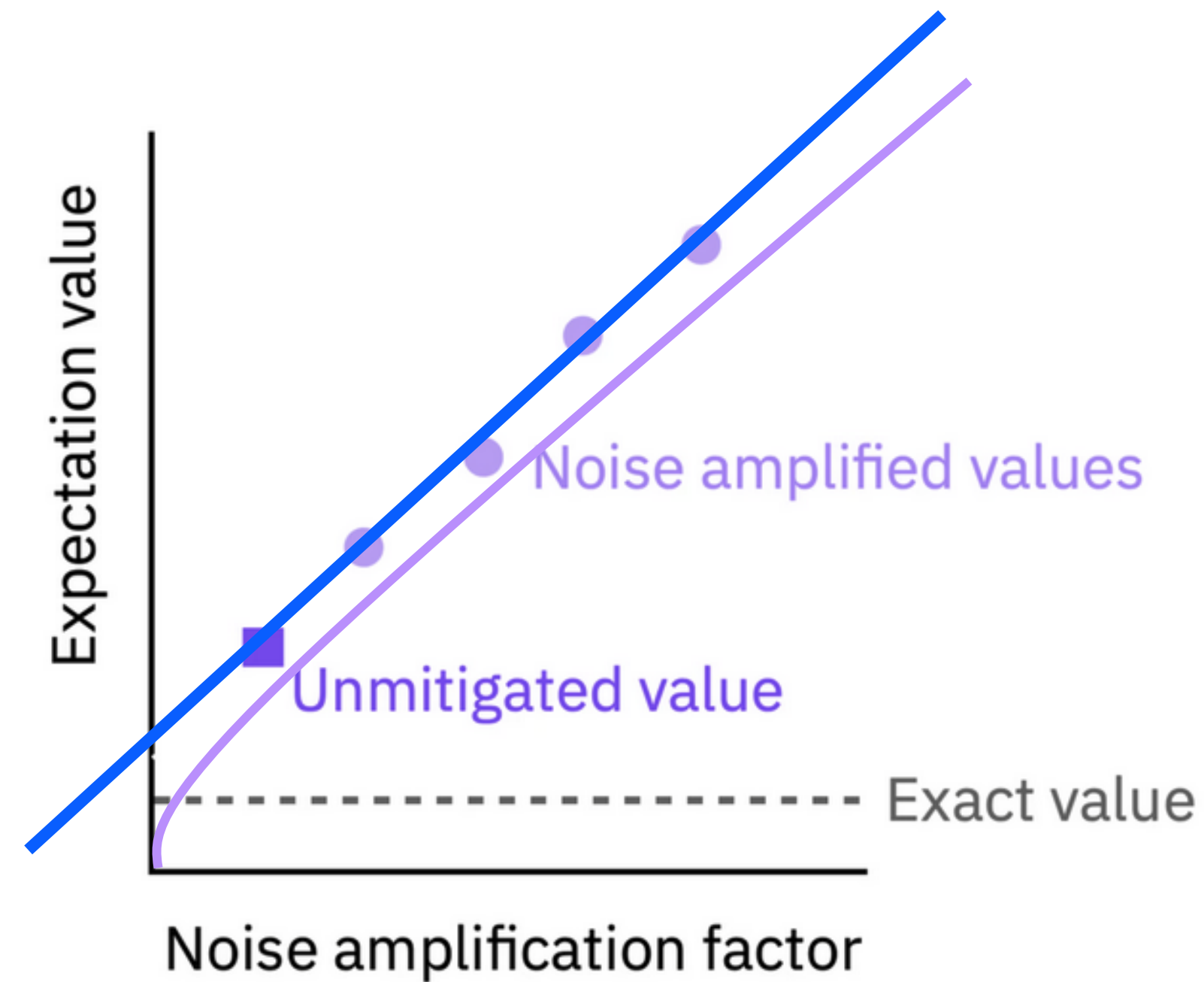
Add noise via sampling Pauli channels



Li & Benjamin PRX (2017)

Noise Extrapolation

- Broad variety of different extrapolators.
- Results can vary depending on the problem.
- Noise learning helps to understand how noise behave.
- Noise can be learned using a Pauli noise model where the errors are described by the **Pauli operators**:
 - I (nothing)
 - X (bit flip)
 - Z (phase flip)
 - Y (bit-and-phase flip)



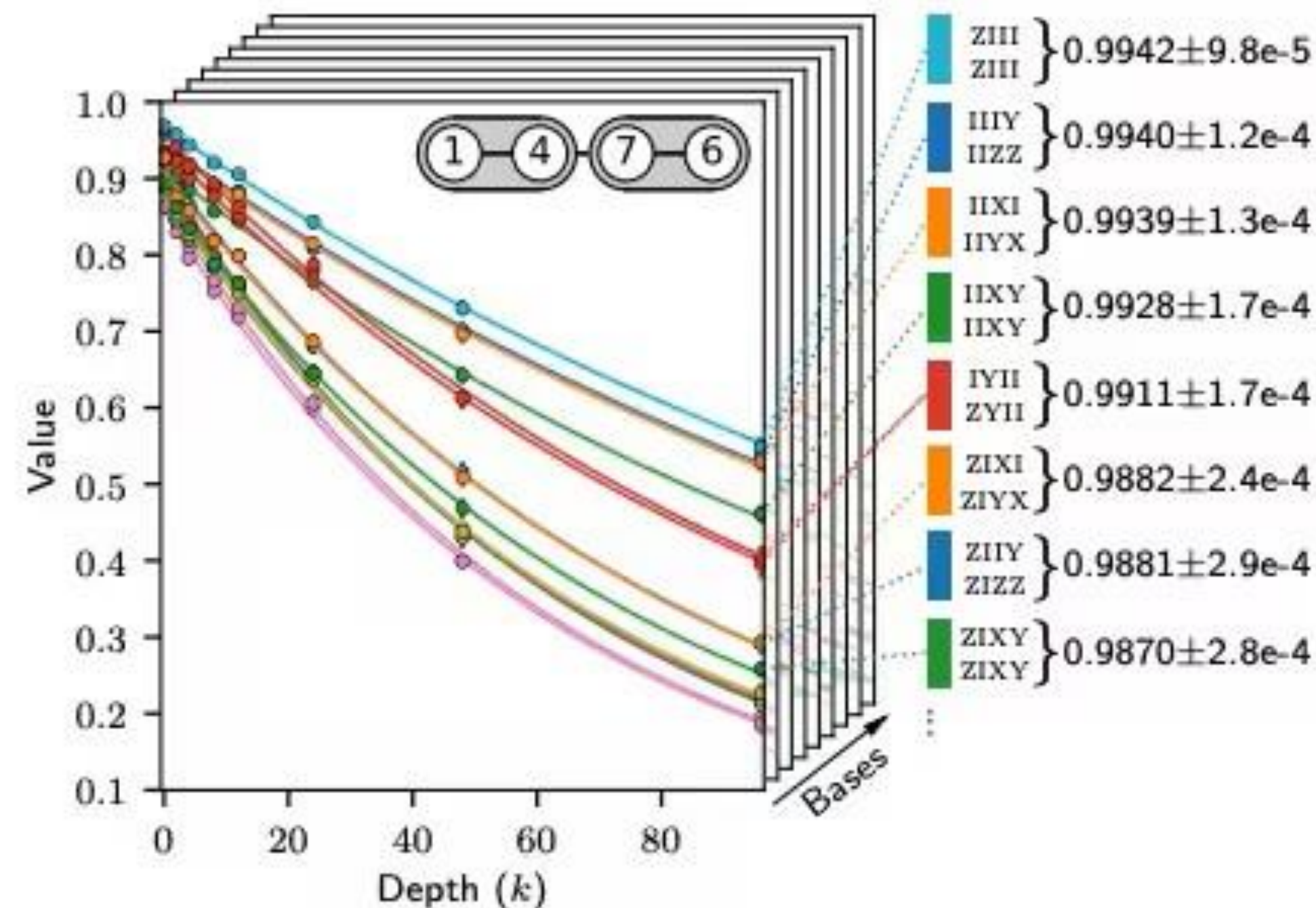
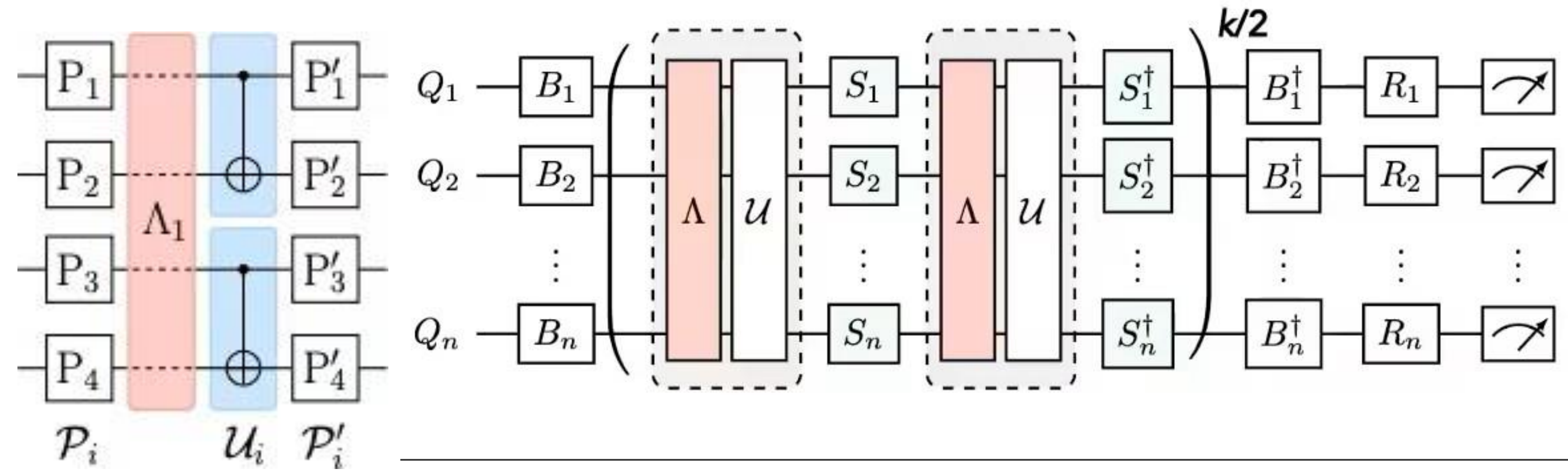
extrapolator

```
Type: UnsetType | Literal['linear', 'exponential', 'double_exponential',  
'polynomial_degree_1', 'polynomial_degree_2', 'polynomial_degree_3',  
'polynomial_degree_4', 'polynomial_degree_5', 'polynomial_degree_6',  
'polynomial_degree_7'] | Sequence[Literal['linear', 'exponential', 'double_exponential',  
'polynomial_degree_1', 'polynomial_degree_2', 'polynomial_degree_3',  
'polynomial_degree_4', 'polynomial_degree_5', 'polynomial_degree_6',  
'polynomial_degree_7']]
```

Default value: Unset

Noise Learning

- [Pauli-twirl](#) the two-qubit gate layers
 - Repeat the identity pairs of layers to learn the noise.
 - Derive a fidelity for each noise channel.
-
- Automatic part of [probabilistic error amplification \(PEA\)](#) for [ZNE](#), but can be done separately.
 - Since only two-qubit gates are used, no relearning needed when parameter changes in a circuit.



Probabilistic Error Cancellation (PEC)

- Similar idea to [probabilistic error amplification \(PEA\)](#)
- The effect of an ideal target circuit can be expressed as a linear combination of noisy circuits that are implementable in practice
- Works for the Estimator primitive

$$\mathcal{O}_{\text{ideal}} = \sum_i \eta_i \mathcal{O}_{\text{noisy},i}$$

Unlike ZNE, it produces bias free expectation values at the cost of increased variance. To account for this, we must take γ^2 times more shots, where:

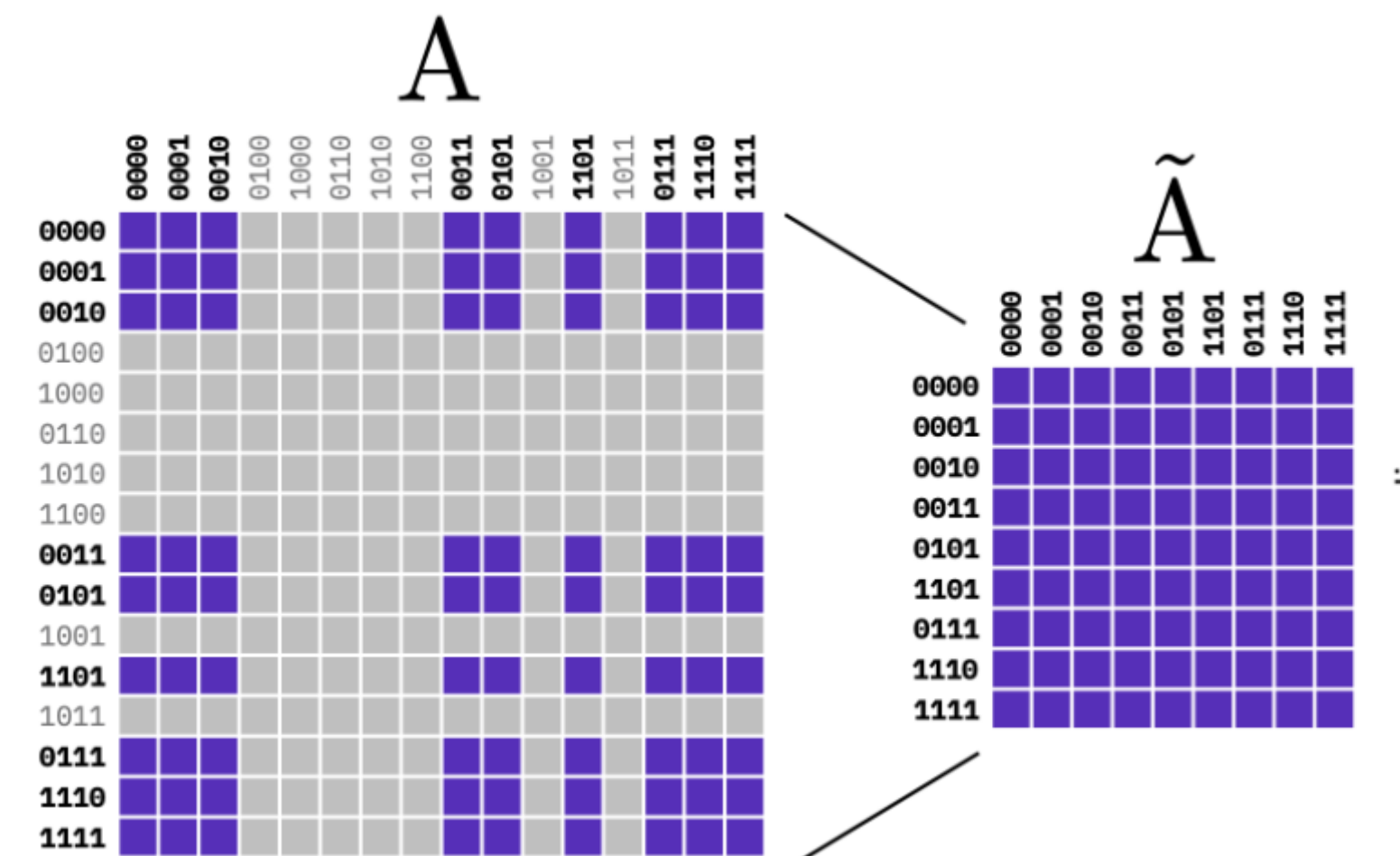
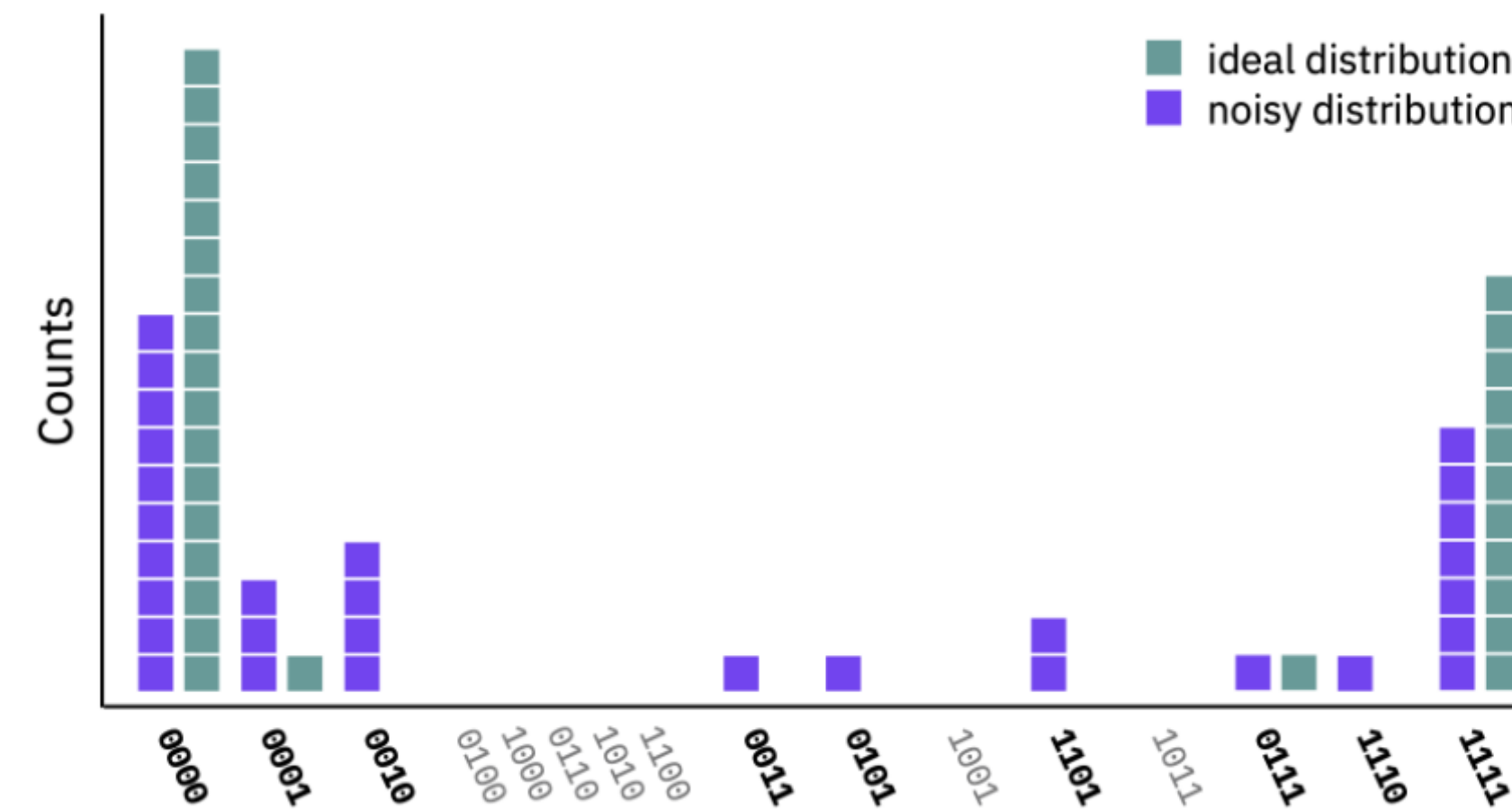
$$\gamma = \sum_i |\eta_i| \geq 1.$$

Caveat: γ scales exponentially with the depth of the circuit

M3 (Qiskit Addon)

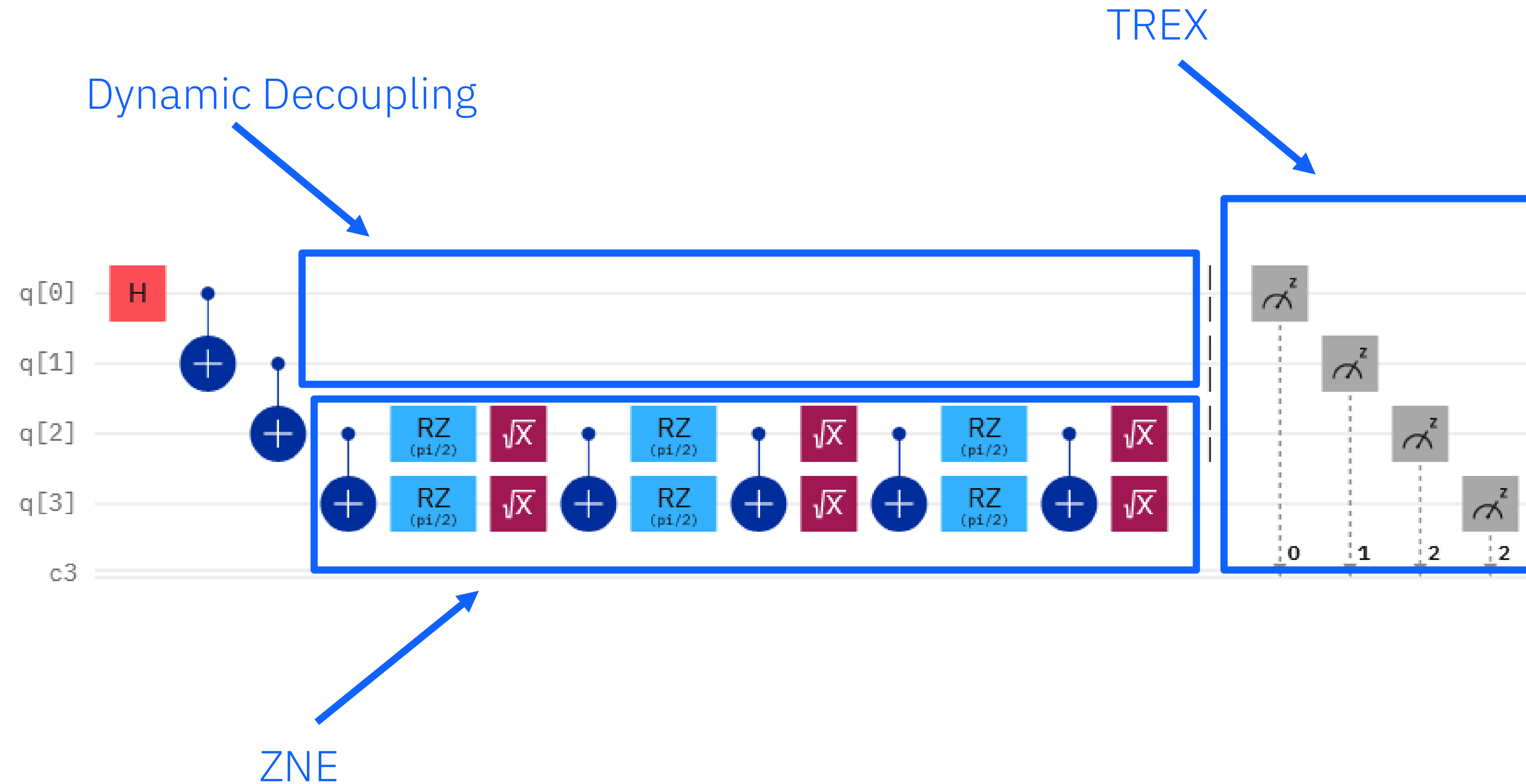
- **M3** works in a reduced subspace defined by the noisy input bitstrings that are to be corrected.
- Because the number of unique bitstrings can be much smaller than the dimensionality of the full multi-qubit Hilbert space, the resulting linear system of equations is nominally much easier to solve.
- Works for the Sampler primitive

Matrix-free Measurement Mitigation (M3)



Recap

- Different types of noise need different suppression and mitigation techniques.
- Different types of techniques can be combined!



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3 Error correction and error detection

Error correction



- **Detects and corrects** errors to **recover** the original quantum state.
- Enables reliable computation **without discarding** results.

Error detection



- **Identifies** whether an error has occurred but does **not correct** it.
- If an error is detected, the faulty result is **discarded**.

Error correction

- Quantum computers are highly susceptible to errors:
 - Unwanted interactions (decoherence)
 - Limited accuracy (gate errors)
- It is widely believed that error correction will be essential for large-scale quantum computing.
- It consists of a quantum computing code to completely correct the errors. In contrast with quantum error mitigation (QEM).

Example: Classical repetition codes

3-bit repetition code

Encoding

$0 \mapsto 000$

$1 \mapsto 111$

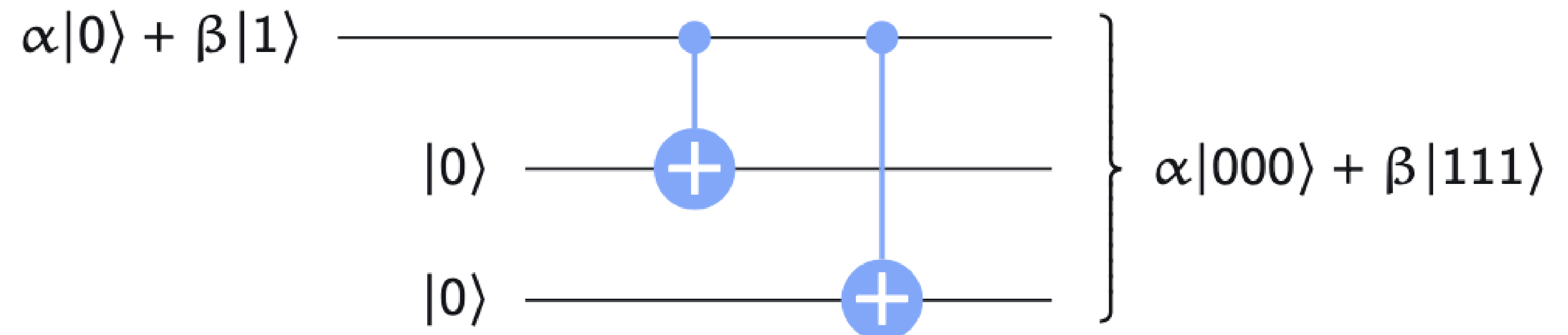
Decoding

$abc \mapsto \text{majority}(a, b, c)$

This code corrects up to one **bit flip** on any of the three bits used for encoding.

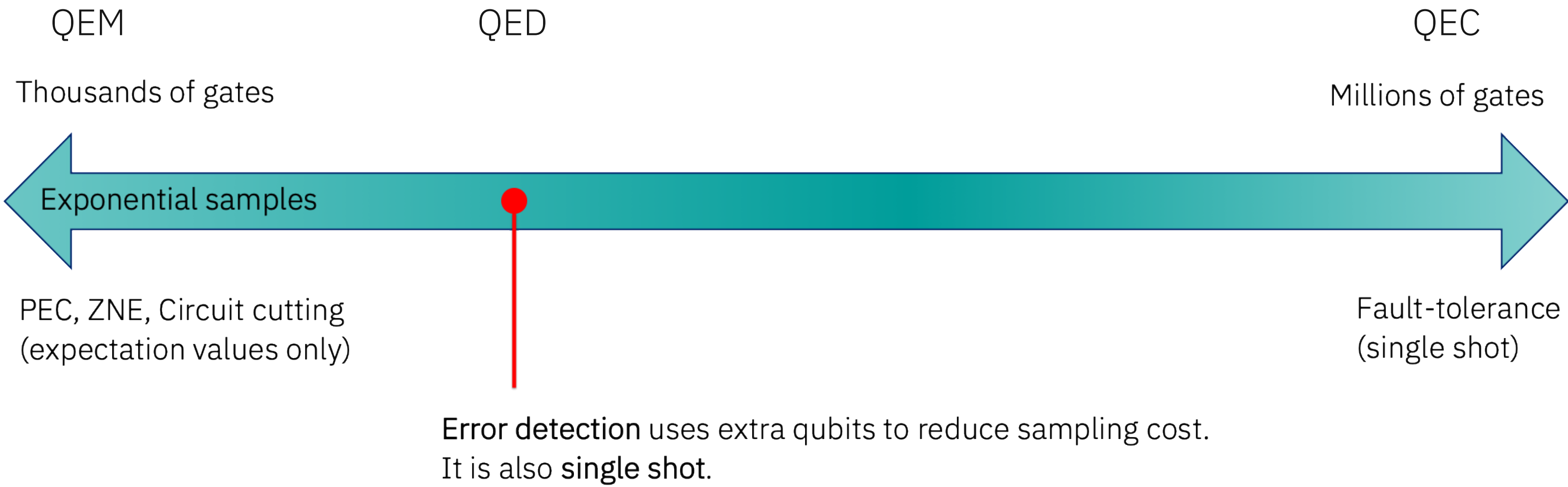
For one qubit the quantum circuit is

$$\alpha|0\rangle + \beta|1\rangle \mapsto \alpha|000\rangle + \beta|111\rangle$$



Error detection

- Measuring ancillary qubits it is possible to detect if an error has occurred.
- It tells you if something went wrong but not how to fix it. In contrast with quantum error correction (QEC).
- Combined with other strategies. Discard runs where errors are detected.

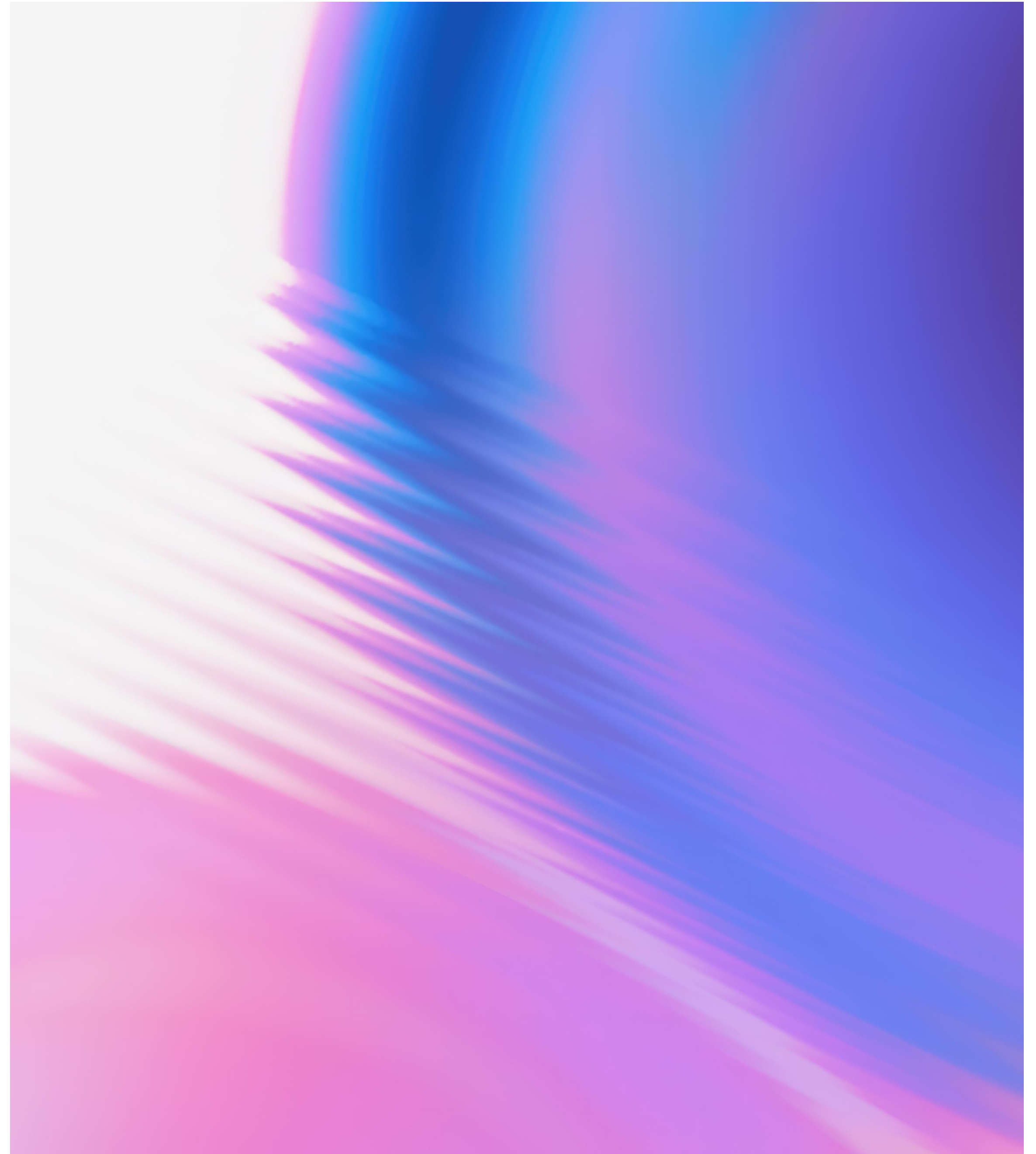


Consider an n -qubit quantum circuit with m gates being executed on a quantum computer with gate error p .

We would like to boost the fidelity from $F = (1 - p)^m$ to $F' = 1 - \epsilon$.

Method	Qubit overhead	Sampling overhead	Single shot
Mitigation (QEM)	-	$\mathcal{O}\left(\left(\frac{1-\epsilon}{(1-p)^m}\right)^4\right)$	×
Detection (QED)	$\mathcal{O}(1)$	$\mathcal{O}\left(\frac{1-\epsilon}{(1-p)^m}\right)$	✓
Correction (QEC)	$\mathcal{O}\left(n \cdot \log_p^2(\epsilon/m)\right)$	-	✓

Fault Tolerance at IBM Quantum



What do we mean by a large-scale fault-tolerant quantum computer?

Large-scale

Hundreds of qubits capable of running hundreds of millions of gates

Fault-tolerant

>100 million operations with broad resistance to failures

What's the difference between error correction and fault tolerance?

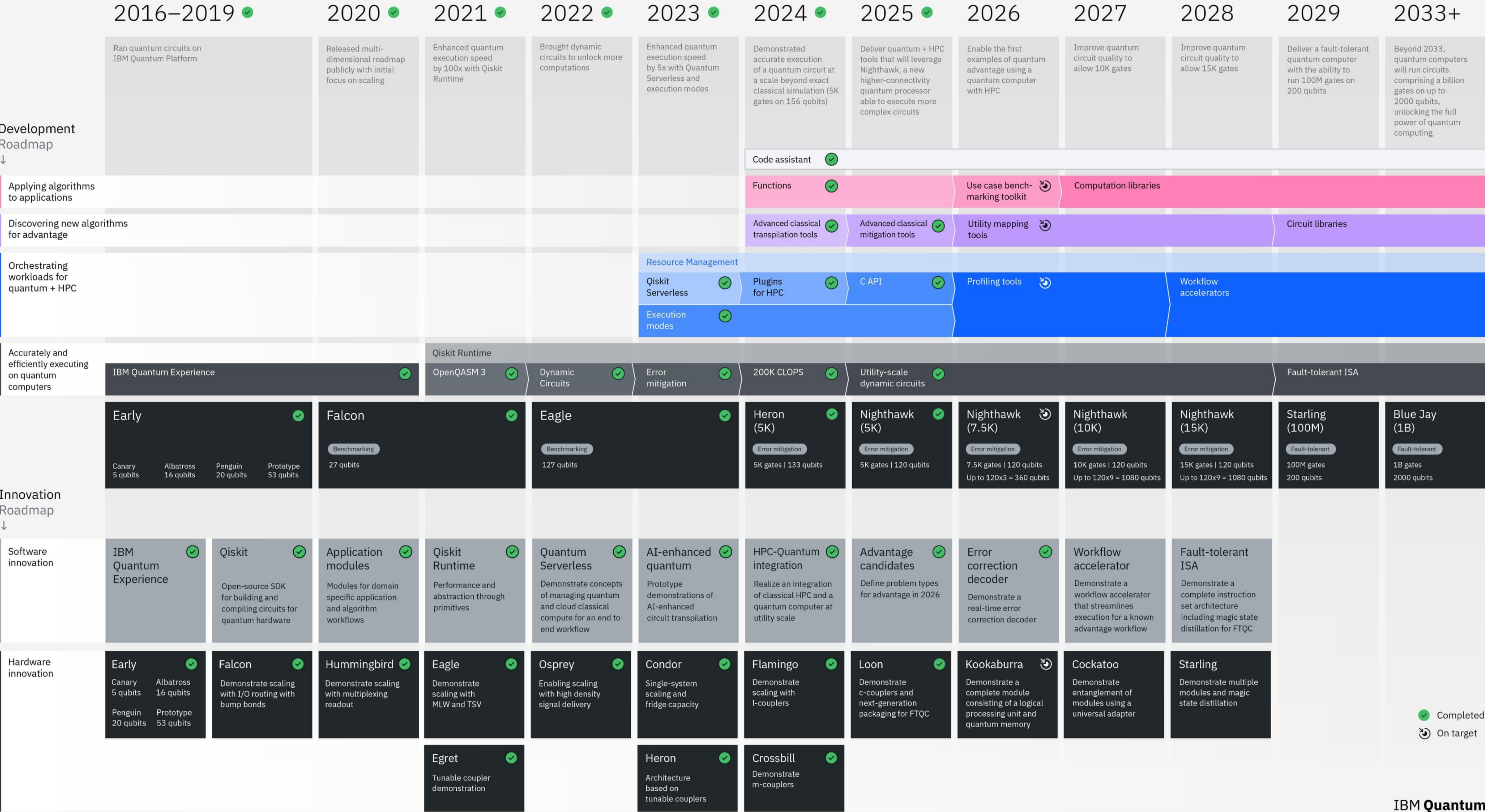
Quantum error correction refers to a family of techniques using quantum error-correcting codes where we encode quantum information into physical qubits so that the information can be detected or corrected by post-selection or with feedback using dynamic circuits. Some techniques are important tests but may not reach the desired scale on their own.

Fault tolerance is about computing capabilities. It applies quantum error-correcting codes so that computations scale efficiently, without incurring an exponential overhead.

IBM is building the first large-scale fault-tolerant quantum computer in the IBM Quantum Data Center in Poughkeepsie, NY.



IBM Quantum Data Center
Poughkeepsie, NY | April 2025



Read the IBM Quantum blog article from June 10th 2025 about the updated roadmap here:

<https://www.ibm.com/quantum/blog/large-scale-ftqc>



Quantum Research

↳ [Blog](#)

How IBM will build the world's first large-scale, fault-tolerant quantum computer

With two new research papers and an updated quantum roadmap, IBM® lays out a clear, rigorous, comprehensive framework for realizing a large-scale, fault-tolerant quantum computer by 2029.

Removing the roadblocks to fault tolerance at scale

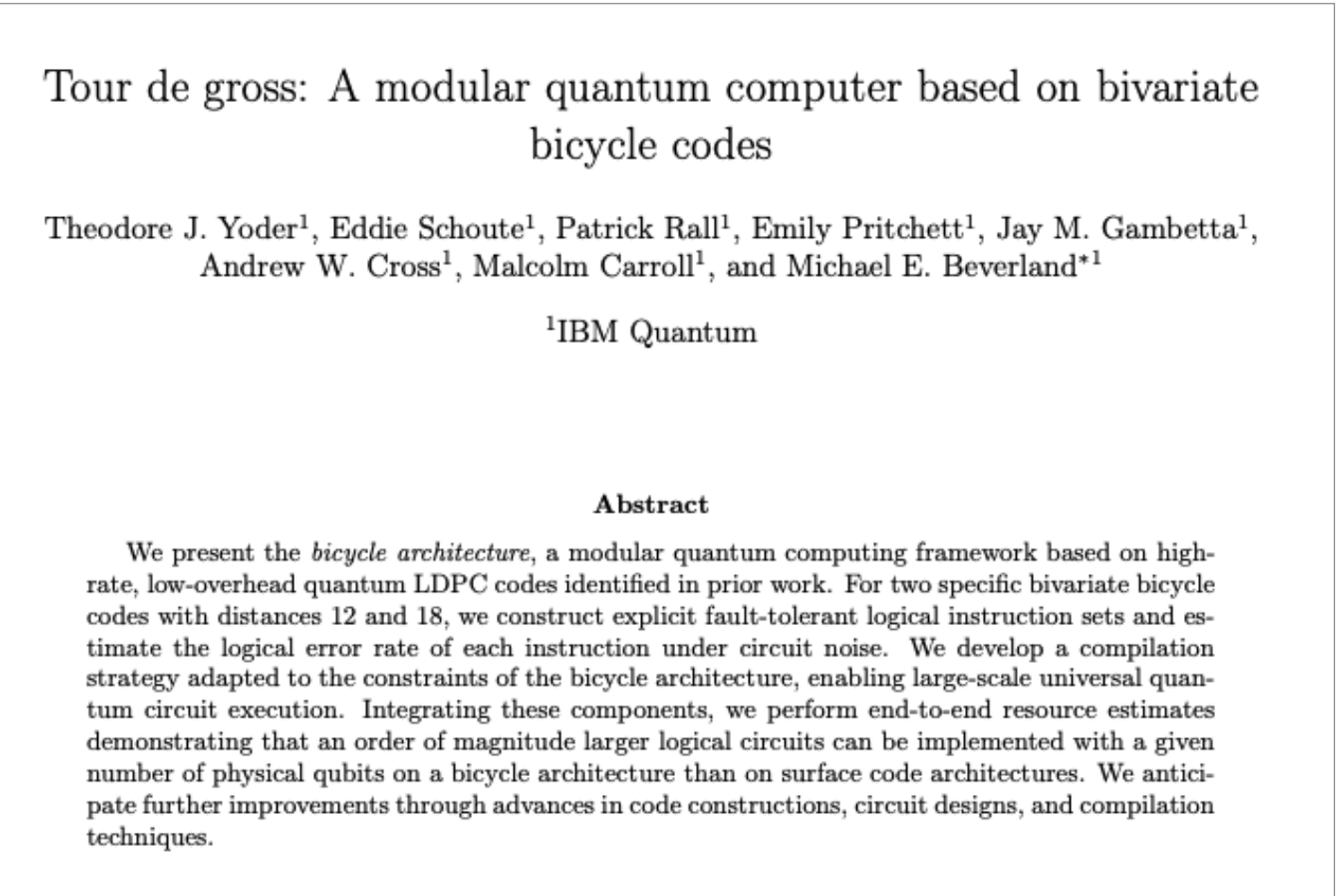
Is there a code that scales efficiently?



Nature 627, 778–782 (2024)

Quantum memory using bivariate bicycle codes requires ~10x fewer physical qubits than the surface code

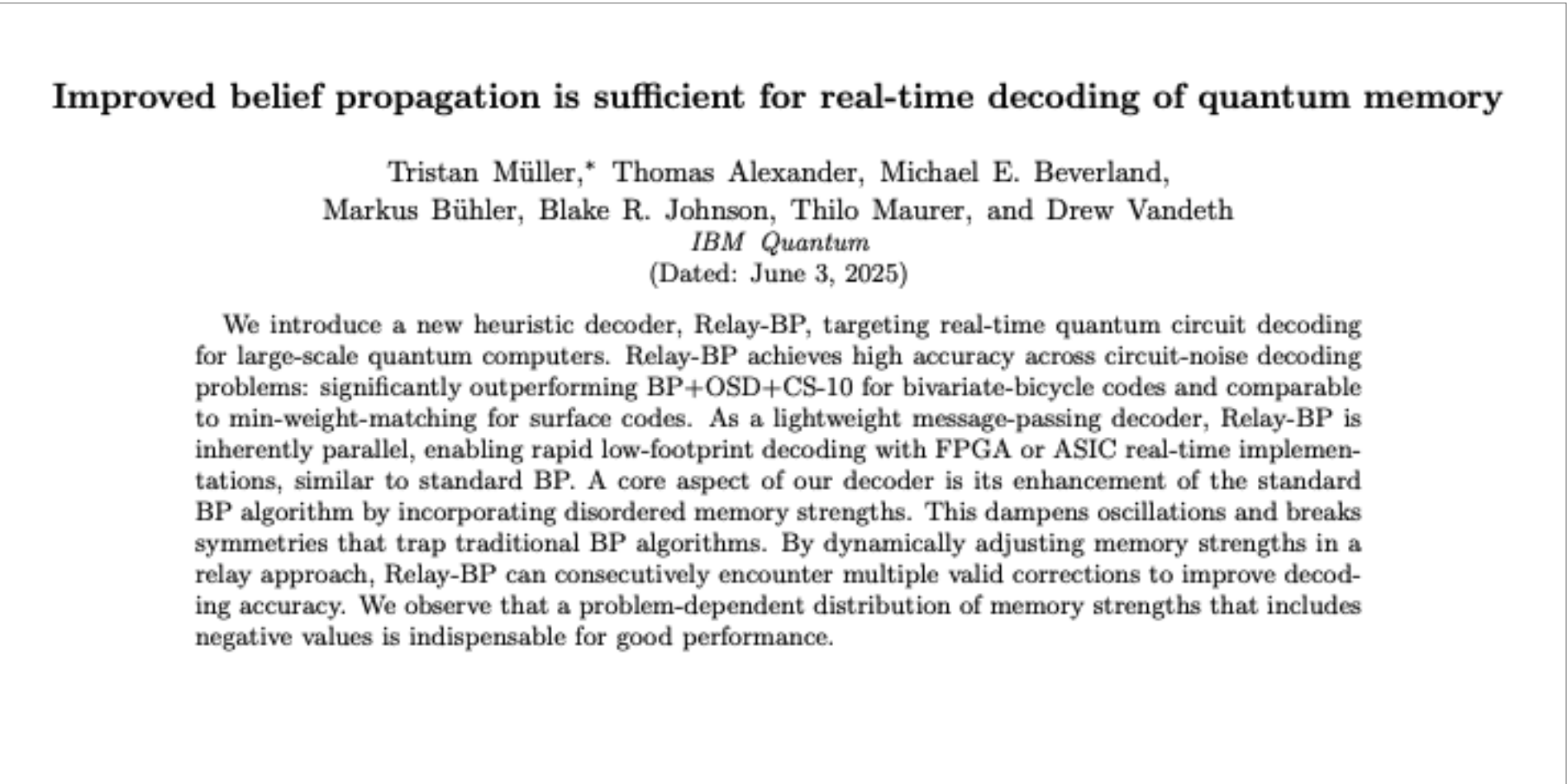
Does this code retain its advantage during computation?



arXiv:2506.03094 (2025)

End-to-end resource estimates demonstrating that bicycle architectures retain an order of magnitude qubit advantage over surface code architectures when implementing large logical circuits

Is real-time decoding possible?

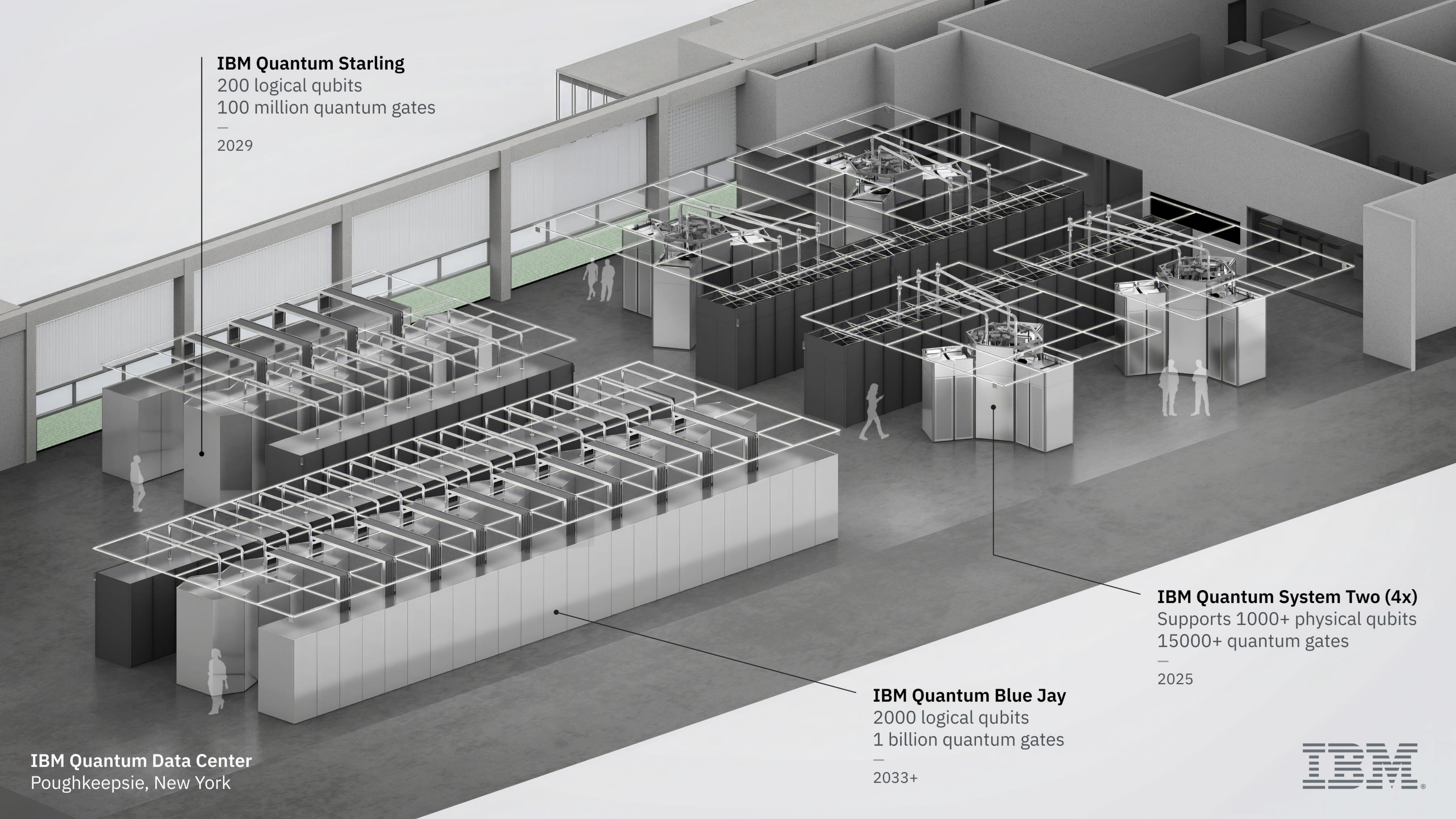


arXiv:2506.01779 (2025)

Demonstrating the feasibility of real-time decoding of qLDPC codes such as bivariate bicycle codes with a modified algorithm (Relay-BP) that eliminates the need for a second-stage decoder.

IBM Quantum Starling

The first large-scale
fault-tolerant
quantum computer



IBM Quantum Starling
200 logical qubits
100 million quantum gates
—
2029

IBM Quantum Data Center
Poughkeepsie, New York

IBM Quantum Blue Jay
2000 logical qubits
1 billion quantum gates
—
2033+

IBM Quantum System Two (4x)
Supports 1000+ physical qubits
15000+ quantum gates
—
2025





Further reading



Courses:

Foundations of quantum error correction course -
<https://quantum.cloud.ibm.com/learning/en/courses/foundations-of-quantum-error-correction>

Others

What's the difference between error suppression, error mitigation, and error correction?

<https://www.ibm.com/quantum/blog/quantum-error-suppression-mitigation-correction>

IBM Quantum development and innovation roadmap

<https://ibm.com/quantum/technology>

How IBM will build the world's first large-scale, fault-tolerant quantum computer

<https://www.ibm.com/quantum/blog/large-scale-ftqc>